Cybernetic Security

Management and monitoring of services and networks

Tomáš Sochor
Dept. of Informatics
Faculty of Business and Economics, Mendel University

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Lecture Agenda

Service monitoring goals
Metrics for server monitoring
Server monitoring tools
Logging tools
Network Monitoring and Discovery

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Service Monitoring Intro

- IT operation goal: Service availability any time necessary.
  - Failure of an IT system/component does not happen “by an act of God”:
    - Every failure roots from causes,
    - And has its symptoms in advance.
  - Every failure causes some losses, e.g.:
    - Loss of reputation (not so reliable organization),
    - Financial loss (e.g. - some customers could have not satisfied their demand with us),
    - Loss of productivity (some employees could not work with an IS).
- Keeping an eye on how every individual IT system/component works is essential:
  - Monitoring
    - By monitoring, we can reveal issues before they cause a real problem/system failure:
      - e.g. by monitoring threshold values for CPU and memory usage, disk capacity, …

Service Monitoring Goals

- Service availability requirement:
  - 24/7 for most services,
    - even in environments with “single shift”:
      - some employees still tend/need to access services in evenings or nights.
- Service parameters:
  - Local:
    - Availability (service running),
    - Resource usage (CPU, memory, disk space, ...) – either physical or virtual.
  - Remote:
    - Availability (service available from remote locations),
    - Latency (response time),
    - Throughput (bytes per second),
    - Other QoS (jitter, max. delay, …).

Server Monitoring Issues

- Is the service available?
  - The process can be running, but it may not be enough:
    - Is the server “alive” remotely (ping)?
    - Is the port “opened” (e.g. listening)?
    - Is it responding promptly?
  - Availability 24/7:
    - Required for most services nowadays.
- How to measure the service availability?

Service Monitoring – How to Measure?

- Metrics for Service monitoring:
  - Local metrics:
    - Virtualization-based:
      - Hypervisor availability,
      - VM state (running/suspended/stopped),
      - Process running,
      - Hardware resources usage (CPU, memory, disk, …)
  - Remote-access based (service availability) metrics:
    - Remote connectivity to the virtual machine:
      - IPv4/IPv6 ping,
      - Port open/responding.
    - Service quality:
      - Application response time (latency),
      - Transmissions rate (for downloading page contents): DON NOT confuse with latency.
      - Other QoS parameters (in specific cases).

Server Monitoring Tools

- Wide variety of commercial and open-source tools:
  - Open-source examples: Prometheus, Nagios, Zabbix.
- Main requirements:
  - Wide choice of metrics:
    - Standard/custom-defined:
      - With alerts when a threshold is exceeded (e.g. too long latency, too high CPU usage).
    - Anomaly detection (based on significant deviation from metric baseline = normal state).
  - Correlation of metrics for debugging an issue:
    - Data from multiple sources (time synchronization!)
      - Servers, switches, firewalls, …
  - Cloud resources support:
    - Cloud services are now widespread.
    - How to measure cloud service availability:
      - What metrics are/are not applicable?

Server Logging

- Monitoring may not be sufficient:
  - Logs should be kept:
    - They can differ in size (what is logged and what is omitted):
      - Event severity,
    - Difference in retention (time period kept).
    - Time synchronization is required.
- Several types of logging is distinguished:
    - Security,
    - Server,
    - Applications.
      - Various retention time,
      - Various severity level kept.

Logging Protocols and Tools

- Syslog:
  - Generic protocol for keeping text log records.
    - Usually in a remote location (syslogd).
    - Common in unix servers.
      - IETF standard (latest RFC 5424 from 2009),
      - Other protocols exist inspired by syslog:
        - E.g. RELP.
- Windows Event Log:
  - Windows-only tool:
      - Frequently used on Windows servers:
        - Better search tools,
        - Difficult to interoperate with other OS platforms
          - e.g. for centralized logging.

Network Monitoring

- SNMP protocol: IETF standard
  - Versions 1, 2c and 3 differing in supported features,
  - Client-server architecture (here, called agent/manager):
    - Agent: monitored device, Manager: single control device.
  - L7 protocol operating above UDP (agent's well-known port 161, manager's for traps 162)
  - GetRequest, SetRequest messages to Agent, Response to Manager
  - Trap: unsolicited message from Agent (triggered by an event on a device).
- MIB (Management Information Base):
  - All SNMP data sent to Manager are stored as variables in MIB.
- Security: plaintext Community string (v1+v2c)
  - Authentication and encryption of messages are supported in version 3.

Management Information Base in SNMP

All SNMP data sent to Manager are stored as variables in MIB:

- Object notation: ASN.1 (ITU standard X.680 as revised in 2021).
  - Each object (variable) is identified with OID string, e.g.
    - 1.3.6.1.4.1.14988.1.1.1.2
      - 1 ISO
      - 1.3 identified-organization,
      - 1.3.6 DoD,
      - 1.3.6.1 internet,
      - 1.3.6.1.4 private,
      - 1.3.6.1.4.1 IANA enterprise numbers,
      - 1.3.6.1.4.1.14988 Mikrotik,
      - …… …1.1.1.2: proprietary structure (device-specific)
    - Mikrotik console: `interface print oid` command

SNMP Manager

Software running typically on a single computer in a network:

- Collecting data from all agents:
  - Identified by the same Community string
- Data are collected from:
  - Responses to GetRequest, SetRequest, … messages
    - Usually sent periodically by the Manager.
  - Traps from agents
    - Unsolicited messages triggered by a predefined event on a device with agent:
      - e.g. CPU usage exceeding 80%

SNMP Agent

Software module available on almost any manageable network device:

- Router,
- Switch,
- IDS/IPS,
- Network firewall.

- Sometimes available also on servers etc.

- SNMP - Legacy protocol:
  - Gradually replaced with newer alternatives.
  - Still kept due to wide support across devices by all vendors.

Network Monitoring – Traffic Flows

- NetFlow:
  - Cisco proprietary protocol since 1996:
    - 3-tier architecture:
      - Flow exporter: the device
      - Flow collector
      - Analyzer app.
    - Supported in some non-cisco hardware.
    - Instead of individual packets, summarizes information into flows:
      - Easier to understand,
      - Less data must be stored.
    - Flow: group of IP packets with identical:
        - IP addresses (source+destination),
        - Ports (source+destination)
        - Protocol number (e.g. 6 form TCP or 1 for ICMP).
  - Various versions:
    - v5 and v9 are the most common:
      - v9 is quite flexible: IPv6 support, MLPLS and many ohters.

Traffic Flow Monitoring

- Alternatives to Netflow:
  - IP Flow Information Export (IPFIX):
    - IETF standard: RFC 7011 – RFC 7015, RFC 5103.
  - sFlow:
    - Sampled flow: random sampling of every n-th packet:
      - Sampling is an alternative to flow aggregation:
        - Less precise but less demanding.
      - Not IETF standard but described in RFC 3176.
  - J-Flow:
    - Juniper proprietary protocol.

RMON, SMON

- Remote Monitoring of Network:
  - Physical and data-link layer monitoring:
    - MIB support: new MIB items are defined.
    - Probes running on devices (routers, switches) send MIB data to SNMP manager.
  - IETF standard: RFC 4502 (v1 RFC 2819).
- SMON: extended RMON with switched networks:
  - RFC 2613.

Network Discovery

- Purpose of Network Discovery:
  - Run on a network infrastructure device (router, switch, firewall, …).
  - Identification of networked devices in the close vicinity of a network device.
    - e.g. nearby routers / switches (directly connected) with:
      - Capabilities:
        - E.g. L3 routing, L2 switching, …
      - Interfaces and their hardware types:
        - GigabitEthernet, Serial, FastEthernet, …
      - Some additional information.
  - Proper function of device discovery depends on running the same protocol on both interconnected devices.

Network Discovery

- CDP (Cisco Discovery Protocol):
  - Proprietary for Cisco devices only:
    - Missing interoperability in heterogeneous infrastructures.
- LLDP (Link-Layer Discovery Protocol):
  - IEEE standard 802.1AB,
  - Interoperability among different switch/router vendors:
    - e.g. Cisco, HP, Dell, Juniper.
    - also with SNMP and MIB: more general approach.

Questions?
Comments?

Tomáš Sochor
Department of Informatics
tomas.sochor@mendelu.cz

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Thank you for your attention

Tomáš Sochor
FBE MendelU
tomas.sochor@mendelu.cz

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